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Lighted surgical instrument

Abstract

A lighted surgical instrument has an instrument body defining a proximal portion, a distal portion, and a mid-portion extending between the proximal portion and the distal portion. The proximal portion includes a base adapted to receive a light source and a linear section defining a first axis, while the distal portion has a linear section defining a second axis and terminating in a functional tip, with the second axis of the distal portion is parallel to but offset from the first axis of the proximal portion. The linear section of the proximal portion may have a cross-sectional shape of a Reuleaux triangle. The instrument body is capable of functioning as a light pipe, such that light introduced into the base of the proximal portion will be transmitted to the functional tip of the instrument by internal reflection.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4043322	12/1976	Robinson	606/160	A61B 1/303
4572180	12/1985	Deenadayalu	600/249	A61F 11/006
5234452	12/1992	Wang-On	606/160	A61B 17/50
5586989	12/1995	Bray, Jr.	606/160	A61B 17/320708
6398793	12/2001	McGuire	606/160	A61B 17/320708
D771815	12/2015	Huttner	D24/151	N/A
10092168	12/2017	Huttner	N/A	A61B 17/062
11166740	12/2020	Schumaier	N/A	A61B 90/35
2008/0092337	12/2007	Gross	81/489	B25G 1/102
2010/0042122	12/2009	Shaw, Jr.	606/162	A61F 11/006
2013/0190647	12/2012	Pahuja	606/162	A61F 11/006
2016/0184046	12/2015	Blain	600/249	A61B 17/1659

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Background/Summary

BACKGROUND OF THE INVENTION

(1) Ear curettage is a commonly performed procedure, usually done for the purposes of removing occluding wax (cerumen) and/or foreign bodies from the ear canal. As an example, approximately 10 million visits to a physician are made annually for complaints of ear pain in children. In a large majority of these patients, occluding cerumen must be removed first to enable the physician to properly examine the ear canal.

(2) Ear curettage has typically been a “blind” procedure. The physician first notes the presence and placement of the occluding cerumen through direct examination of the ear canal using an otoscope. Then, using a non-lighted ear curette, the physician blindly scrapes the ear canal in an attempt to remove the obstruction. This can be a painful process, and one that is not always successful in removing the occluding wax.

(3) An improvement to this procedure occurred with the introduction of the lighted ear curette. This

device used a clear plastic curette as a light pipe to conduct light from an attached light source into the ear canal. Using a lighted ear curette the physician could now illuminate the ear canal to enable visualization of the occluding cerumen and allow its removal under direct observation.

(4) The lighted ear curette still has deficiencies, however. The lighted ear curette is designed as having a single linear axis, resulting in a straight light pipe. This means that light entering the rearward end of the curette travels by total internal reflection in an axial direction until it emits from the forward tip of the curette. It can be seen in FIG. 1 that the light-path axis of the curette and the visual (eye line) axis are not the same. Depending on the user and how they hold the curette, the visual axis may be 15 degrees or greater above the light-path axis when viewed in the frontal plane. This can cause problems when trying to visualize cerumen in a long, narrow tube such as the ear canal, as the hand and fingers may obstruct the view. This problem is accentuated when trying to visualize the narrower ear canal of children. Although light may illuminate the distal reaches of the ear canal, the angular deviation between the visual axis and light-path axis will limit the depth to which adequate visualization can occur.

(5) A second deficiency with the current lighted ear curette is that there is no region designed for natural finger grip placement. As curettage is often a delicate procedure performed on very young children, it is important that the user have a solid, natural grasp on the curette.

SUMMARY OF THE INVENTION

(6) The invention is a lighted surgical instrument has an instrument body defining a proximal portion, a distal portion, and a mid-portion extending between the proximal portion and the distal portion. The proximal portion includes a base adapted to receive a light source and a linear section defining a first axis, while the distal portion has a linear section defining a second axis and terminating in a functional tip, with the second axis of the distal portion is parallel to but offset from the first axis of the proximal portion. The linear section of the proximal portion may have a cross-sectional shape of a Reuleaux triangle. The instrument body is capable of functioning as a light pipe, such that light introduced into the base of the proximal portion will be transmitted to the functional tip of the instrument by internal reflection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The advantages of the invention will become readily apparent to those skilled in the art from the following detailed description of various embodiments when considered in the light of the accompanying drawings, in which:

(2) FIG. 1 is a somewhat schematic view of a prior art ear curette in use, depicting the visual path of the user vs the axis of the prior art ear curette;

(3) FIG. 2 is a side view of a preferred embodiment of a lighted ear curette in accordance with the invention;

(4) FIG. 3 is an enlarged, cross-sectional view through lines 3-3 of FIG. 2;

(5) FIG. 4 is a side view of the ear curette of FIG. 2 with light source attached to the distal end thereof; and

(6) FIG. 5 is a somewhat schematic view of the ear curette of FIG. 2 in use, depicting the visual path of the user vs the axis of the ear curette.

DETAILED DESCRIPTION OF THE INVENTION

(7) It is to be understood that the specific devices and processes illustrated in the attached drawings and described in the following description are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein should not be considered as limiting, unless the claims expressly state otherwise.

(8) The invention is an improved, lighted surgical instrument **10**, especially an improved lighted ear curette. In certain embodiments, the invention addresses deficiencies in the conventional lighted ear curette by introducing a lighted curette wherein the visual axis and light-path axis are identical, providing full illumination and enhanced visualization of obstructions in the ear canal, and/or the invention addresses other deficiencies in the conventional lighted ear curette by adding a unique triangular shape to the proximal portion of the curette for an enhanced, ergonomic grip.

(9) The lighted surgical instrument of the invention includes an instrument body **12**. The instrument body **12** is capable of functioning as a light pipe, such that light introduced into one end of the body will be transmitted therethrough and exit the other end of the body. In the embodiment of the invention illustrated in FIG. 2, the instrument body **12** is a curette body having an off-set design defining three portions of the curette. The instrument body **12** defines a proximal portion **14**, a distal portion **16**, and a mid-portion **18** extending between the proximal portion **14** and the distal portion **16**.

(10) The proximal portion **14** of the body **12** includes a rearward base **20** adapted to attach to and receive a light source, and a linear section **22** defining a first axis and acting as a finger-grip portion for the user of the instrument body **12**. A preferred cross-sectional shape of the linear section **22** is shown in FIG. 3. This portion of the curette has a roughly triangular shape in what is known as a "Reuleaux triangle." A Reuleaux triangle is a shape having a boundary that is a curve of constant width. The Reuleaux triangular cross-sectional shape of the linear section **22** of the proximal portion **12** provides a natural, ergonomic finger grip for the device, while still retaining the light pipe properties of the instrument body **12**. In a further preferred aspect, the Reuleaux triangular cross-sectional shape extends the entire length from the base **20** to the start of the mid-portion **18**, allowing proper finger placement by users of varying hand sizes, large and small.

(11) The distal portion **16** of the instrument body **12** has a linear section **24** defining a second axis and terminating in a functional tip **26**. As a result of the angles between the proximal portion **14** and the mid-portion **18**, and then again between the mid-portion **18** and the distal portion **16**, the second axis defined by the distal portion **16** is parallel to but offset from the first axis defined by the proximal portion **14**. Thus, the noted configuration of the portions of the instrument body **12** raises the visual axis of the distal portion **16**/functional tip **26** above the linear axis of the proximal portion **14**.

(12) In certain preferred embodiments, at least a portion of the linear section **24** of the distal portion **16** tapers towards the functional tip **26**. In further embodiments, at least a portion of the linear section **24** of the distal portion **16** is provided with a circular cross-sectional shape.

(13) The working or functional tip **26** of the instrument body **12** may have any of multiple suitable configurations. In the illustrated embodiment, the lighted surgical instrument **10** is a lighted ear curette and the functional tip **26** is configured as a loop. In certain preferred embodiments, the linear section **24** of the distal portion **16** is provided with a plurality of visible index marks **28** to designate depth or position of the functional tip, for example within the ear canal of a patient.

(14) As shown in FIG. 4, a light source **30** may be attached to the base **20** of the instrument body **12**. Preferably, the light source **30** is an LED light contained within a housing that is selectively attachable to the base **20**, such as by a threaded connection, a bayonet-type connection, or the like.

(15) In a preferred embodiment, the entire instrument/curette body **12** is formed of unitary molded construction and is comprised of a clear, light-conducting plastic. A transparent polycarbonate is especially preferred. Preferably, the instrument body exhibits less than 5% haze, and 95% or better clarity. When light is introduced into the rearward face of the base **20** of the proximal portion **14** of the body **12**, the light is transmitted via total internal reflection to the functional tip **26** of the curette, where it is emitted. Preferably, the light introduced into the base **20** will be transmitted to the functional tip **26** of the instrument essentially in its entirety by internal reflection.

(16) FIG. 5 illustrates an embodiment of the lighted curette of the invention in use. As shown in the drawing, the visual axis (defined as a line running from the user's eye line to the distal working tip

of the curette) is roughly identical with the linear axis of the distal portion of the curette. This allows the user to visualize obstructions deeper into the ear canal of the patient than previously possible using an ear curette with only a single, straight light-pipe.

(17) In accordance with the provisions of the patent statutes, the invention has been described in what is considered to represent its preferred embodiments. However, it should be noted that the invention could be practiced otherwise than as specifically illustrated and described without departing from its spirit or scope.

Claims

1. A lighted surgical instrument comprising an instrument body defining a proximal portion, a distal portion, and a mid-portion extending between the proximal portion and the distal portion; wherein the proximal portion is comprised of a base adapted to receive a light source and a linear section defining a first axis; wherein the distal portion is comprised of a linear section defining a second axis and terminating in a functional tip, and wherein at least a portion of the linear section of the distal portion is both solid in cross section and tapers towards the functional tip; wherein the second axis of the distal portion is parallel to but offset from the first axis of the proximal portion; wherein the second axis of the distal portion is co-linear with the visual axis of curettage, as defined by an imaginary line connecting the eye line of a user with the linear axis of an ear canal of a patient; and wherein the instrument body is capable of functioning as a light pipe, such that light introduced into the base of the proximal portion will be transmitted to the functional tip of the instrument by internal reflection.
2. The lighted surgical instrument of claim 1, wherein at least a portion of the linear section of the proximal portion has a cross-sectional shape of a Reuleaux triangle.
3. The lighted surgical instrument of claim 2, wherein the linear section of the proximal portion has a cross-sectional shape of a Reuleaux triangle along its entire length from the base to the mid-portion of the instrument body.
4. The lighted surgical instrument of claim 1, wherein the instrument body is formed of a single unitary piece of molded plastic.
5. The lighted surgical instrument of claim 1, wherein the instrument body is comprised of a transparent polycarbonate.
6. The lighted surgical instrument of claim 1, further comprising a light source attached to the base of the instrument body and configured to introduce light into the base of the proximal portion of the instrument body so that the light is transmitted through the instrument body and is emitted from the functional tip.
7. The lighted surgical instrument of claim 6, wherein the light source is an LED light source.
8. The lighted surgical instrument of claim 1 wherein at least a portion of the linear section of the distal portion has a circular cross-sectional shape.
9. The lighted surgical instrument of claim 1, wherein the instrument body exhibits less than 5% haze.
10. The lighted surgical instrument of claim 1, wherein the instrument body exhibits 95% or better clarity.
11. The lighted surgical instrument of claim 1, wherein the functional tip is comprised of a loop for ear curettage.
12. The lighted surgical instrument of claim 1, wherein the linear section of the distal portion includes a plurality of visible index marks to designate depth or position of the functional tip.
13. A lighted surgical instrument comprising an instrument body formed of a single unitary piece of molded plastic and defining a proximal portion, a distal portion, and a mid-portion extending between the proximal portion and the distal portion; wherein the proximal portion is comprised of a base adapted to receive a light source and a linear section defining a first axis, with at least a

portion of the linear section of the proximal portion having a cross-sectional shape of a Reuleaux triangle; wherein the distal portion is comprised of a linear section defining a second axis and terminating in a functional tip, and wherein at least a portion of the linear section of the distal portion is both solid in cross section and tapers towards the functional tip; wherein the second axis of the distal portion is parallel to but offset from the first axis of the proximal portion; wherein the second axis of the distal portion is co-linear with the visual axis of curettage, as defined by an imaginary line connecting the eye line of a user with the linear axis of an ear canal of a patient; and wherein the instrument body is capable of functioning as a light pipe, such that light introduced into the base of the proximal portion will be transmitted to the functional tip of the instrument by internal reflection.

14. A lighted surgical instrument comprising an instrument body defining a proximal portion, a distal portion, and a mid-portion extending between the proximal portion and the distal portion; wherein the proximal portion is comprised of a base adapted to receive a light source and a linear section defining a first axis; wherein the distal portion is comprised of a linear section defining a second axis and terminating in a functional tip, and wherein at least a portion of the linear section of the distal portion is both circular in cross section and tapers towards the functional tip; wherein the second axis of the distal portion is parallel to but offset from the first axis of the proximal portion; wherein the second axis of the distal portion is co-linear with the visual axis of curettage, as defined by an imaginary line connecting the eye line of a user with the Page 6 linear axis of an ear canal of a patient; and wherein the instrument body is capable of functioning as a light pipe, such that light introduced into the base of the proximal portion will be transmitted to the functional tip of the instrument by internal reflection.
